

Nuzaer Omar

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Research Interests

Adversarial Attacks & Defenses in LLMs, Unsupervised Learning in LLMs, Robust and Reliable LLMs, Black-Box and Zero-Access LLM Security, Context-Aware & Phonetic Perturbations in LLM, Signal Processing

Education

Aug 2022 - Jul 2027 (expected)	■ PhD, Computer Science Department Missouri University of Science & Technology, Supervisor: Dr. Sanjay Madria, Result: <i>CGPA 4.00 out of 4.00, 63 credits</i>
2016 – 2021	■ BSc., Electrical & Electronic Engineering Department Chittagong University of Engineering & Technology. Result: <i>CGPA 3.69 out of 4.00</i>

Employment History

Jun 2026 – Present	■ Research Intern , Argonne National Laboratory, Lemont, IL, USA.
2022 - current	■ Graduate Research Assistant , Wireless to Cloud Computing Lab, Missouri University of Science & Technology,
2021 - 2022	■ Lecturer , Port City International University, Bangladesh.

Research Publications

Conference Proceedings

- 1 Omar, N., Jewel, M. R., & Madria, S. (2026). Thermal Mirage: Semantic Adversarial Attack via Physics-Informed GAN-Guided Diffusion. In *Proceedings of the 34th acm sigspatial international conference on advances in geographic information systems*. Under Communication, Riverside, CA, USA.
- 2 Omar, N., & Madria, S. (2026a). From Perturbation to Reasoning: Knowledge-Guided Multi-Stage Realistic Adversarial Attacks on LLMs. In *Proceedings of the 2026 conference on language modeling*. Under Communication, San Francisco, USA.
- 3 Omar, N., & Madria, S. (2026b). SCALE: Semantic Constraint Aware Lexicon Embedding for Efficient LLM Guided Text Clustering. In *Proceedings of the 2026 conference on empirical methods in natural language processing*. Under Communication, Budapest, Hungary.
- 4 Nuzaer Omar, S. M., Ademola Adesokan. (2025). Leveraging Pre-Trained Language Models for Realistic Adversarial Attacks. In *2025 ieee international conference on big data*.
- 5 Nuzaer Omar, M. U., M. Dey. (2020). Detection of Myocardial Infarction from ECG Signal Through Combining CNN and Bi-LSTM. In *2020 11th international conference on electrical and computer engineering (ICECE)*.
🔗 doi:10.1109/ICECE51571.2020.9393090

Journal Articles

- 1 Nuzaer Omar, M. U., M. Dey. (2021). Temporal Feature-Based Classification into Myocardial Infarction and other CVDs Merging CNN and Bi-LSTM from ECG signal. *IEEE Sensors Journal*, 21, 21688–21695.
🔗 doi:10.1109/JSEN.2021.3079241

Research & Technical Skills

Programming	■ Python, MATLAB, C/C++, R, Verilog, Assembly(x86)
Tools & Frameworks	■ Pytorch, Tensorflow, HuggingFace, Scikit-learn, Openattack, Pandas, OpenCV, spaCy, NLTK, Ollama, vLLM, PySpark

Research & Technical Skills (continued)

Databases	SQL, MySQL, MongoDB, Redis
Softwares/ Research Tools	LaTeX, Overleaf, Git, Docker, Slurm, GNS3, Proteus, Cadence Virtuoso, Pspice
Machine Learning	CNNs, Autoencoders/VAEs, Contrastive Learning, Representation Learning, YOLOv5/8, Clustering
NLP/LLMs	Transformer Models (BERT, T5, GPT, BART, LLaMA), Adversarial Attacks & Defenses, Prompt Engineering, Decoding Strategies, Semantic Similarity (SBERT), Text Generation, Topic Modeling, Phonetic & Lexical Perturbations
Optimization & Modeling	Loss Design (Contrastive, Reconstruction, Triplet, and KL), Hyperparameter Tuning, Model Calibration

Research Experience

Modular Feedback-Optimized Black-Box LLM Attack Framework:

- Designed a training free black-box adversarial text attack system integrating MLM-based perturbations, phonetic perturbations, and PLL-ranked candidate substitutions.
- Developed an iterative feedback optimization loop that refines adversarial candidates to maximize attack success while preserving fluency, perplexity, and semantic similarity constraints, improving attack success by up to 15% with similarity above 80%.

Unsupervised Re-labeling & Robustness Enhancement Framework:

- Built an unsupervised pipeline combining Sim-Text keyword attribution, Seq2Seq autoencoding, unsupervised contrastive representation learning, and KL-divergence clustering.
- Corrected mislabeled neutral samples, boosting F1 by 10% and reducing adversarial attack success by 14% across multiple LLM classifiers.

Physics Informed Semantic Attacks on Thermal Object Detectors:

- Developed a physics informed adversarial framework for thermal object detectors using class aware GAN thermal priors, diffusion guided object perturbation, and distance aware context manipulation.
- Evaluated the attack on FLIR and the proposed MineThermal dataset, achieving 76% ASR against YOLOv11 with high perceptual realism, including SSIM = 0.850 and LPIPS = 0.256, while demonstrating transferability across Faster R-CNN, YOLOv11, and RT-DETR.

Feature-Driven DL Framework for Myocardial Infarction Detection:

- Developed a multi-class MI detection framework using 21 temporal ECG features from fiducial-point extraction. This mitigated redundancy and class imbalance, achieving AUC = 99.25% and F1 = 98.86%, outperforming waveform-based baselines.
- Proposed novel temporal feature extraction algorithms for robust temporal feature extraction to ensure reliable and clinically meaningful detection under imbalanced conditions.

Relevant Graduate Courses

Advanced Topics in Artificial Intelligence | Clustering Algorithms | Machine Learning in Computer Vision | Analysis of Algorithms | Regression Analysis | Network Performance Analysis | Applied Social Network Analysis.

Project Highlights

- Chess:** Implemented a chess program from scratch with single and two-player modes using optimized search and data structures for fast move generation.
- NN:** Built a from-scratch binary classifier implementing custom initialization, loss/gradient calculations, and gradient-descent training.
- Developed a YOLOv5-based object detection pipeline on the FLIR thermal dataset, improving robustness in limited visibility environments.

Project Highlights (continued)

- 📌 Developed a semantic segmentation model on the CARLA simulator dataset for pixel-level road scene parsing in autonomous driving.

Awards and Mentorship

- 2022 - 2026 📌 **Kummer Innovation and Entrepreneurship Doctoral Fellows Program**
Missouri University of Science & Technology.
- 2025 📌 **NSF Local I-CORPS, Great Lakes Region**
- 2022 📌 Mentored a student team at PCIU that achieved a Top-20 placement in the national "Mujib 100 Idea Contest."
- 📌 Reviewer in conferences like ICDM 2025, IEEE Big Data 2024-2025, ECML 2025, etc.